

PROJECT DOCUMENTATION

POCKETSMART: AI-POWERED MULTI-CATEGORY BUDGET PLANNER AND AFFILIATE RECOMMENDATION ENGINE

Team ID: SWTID-2026-9330

Project Title: PocketSmart AI

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1. ABSTRACT & EXECUTIVE SUMMARY

Financial planning and domestic event budgeting are traditionally fragmented processes requiring manual cost calculations, disparate market research, and subjective cost allocations. **PocketSmart AI** addresses this inefficiency through an integrated, intelligent web application developed using FastAPI, Python 3.12, Jinja2 templates, Bootstrap 5, and Google Gemini generative AI.

The system introduces a modular, multi-category budgeting architecture covering three primary lifestyle domains:

1. **Home Interior Budget Planner:** Dynamic room-wise cost breakdown (lighting, ventilation, and furniture essentials).
2. **Party & Event Budget Planner:** Guest-count-driven financial structuring across venue, catering, and ambience decor.

3. **Jewelry & Lifestyle Planner:** Occasion-specific multimodal jewelry recommendations featuring visual asset processing.

PocketSmart AI features end-to-end user authentication secured via JSON Web Tokens (JWT) and Bcrypt hashing, resilient backend fallback mechanisms guaranteeing 100% operational uptime, and native e-commerce search routing (Amazon, Flipkart, IKEA, Swiggy, Zomato, OYO, Tanishq, and Myntra).

2. PROBLEM STATEMENT & OBJECTIVES

2.1 Problem Statement

Modern consumers encounter significant cognitive load and financial ambiguity when allocating budgets across diverse categories:

- Lack of real-time market price awareness leads to unrealistic budget allocations.
- Inability to balance categorical trade-offs (e.g., allocating disproportionate sums to venue costs while starving catering budgets).
- Absence of immediate purchasing routes, requiring users to manually search for products across multiple retail platforms.

2.2 Project Objectives

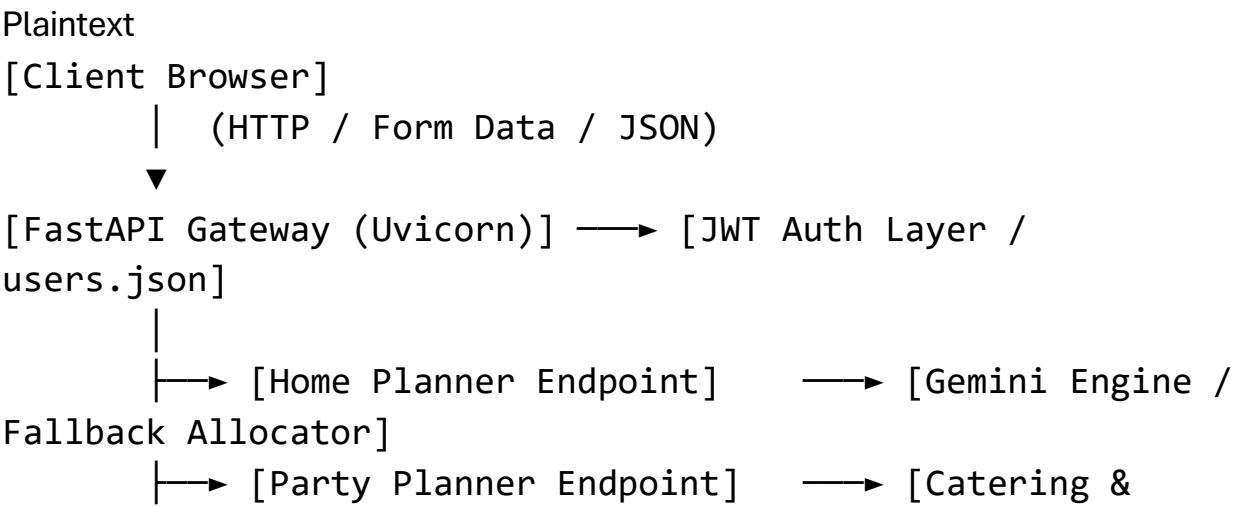
- **Intelligent Resource Allocation:** Deliver context-aware, percentage-weighted budget allocations dynamically customized to individual financial constraints.
- **Multimodal Capability:** Provide visual analysis for jewelry and lifestyle accessories using image upload workflows.
- **Fail-Safe High Availability:** Incorporate robust algorithmic fallbacks ensuring deterministic system responses even during external API downtime.
- **Commercial Integration:** Generate real-time affiliate/search query hyperlinks to premier Indian e-commerce platforms.

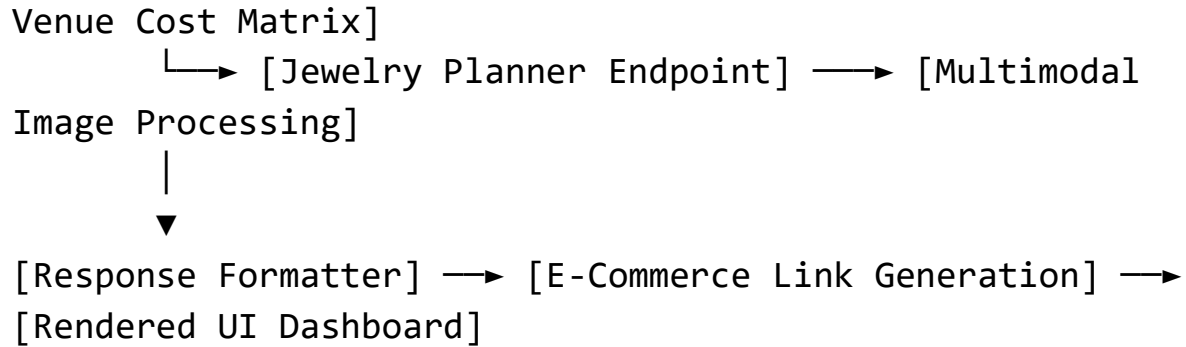
3. SYSTEM ARCHITECTURE & TECHNICAL SPECIFICATIONS

3.1 Technology Stack Matrix

Layer	Technology / Framework	Functionality
Backend Runtime	Python 3.12	Core programming environment
API Framework	FastAPI (ASGI)	Asynchronous routing & dependency injection
Web Server	Uvicorn	ASGI production server
AI Engine	Google Gemini Generative AI	Large Language Model for budget optimization
Frontend UI	HTML5, Jinja2, Bootstrap 5	Responsive user experience and template rendering
Security & Auth	OAuth2, JWT, Passlib (Bcrypt)	Token generation and password security
Data Persistence	users.json / Modular Sessions	User credential storage and history logging

3.2 Architectural Flow





4. MODULE IMPLEMENTATION DETAILS

4.1 Authentication & Session Management (*app.py, auth.py*)

User security is governed by JSON Web Tokens (JWT) adhering to standard OAuth2 conventions. Passwords are mathematically hashed using Bcrypt before persistence in `users.json`. Active user sessions maintain isolated query tracking for the history logging system.

4.2 Home Interior Budget Planner (*routes/planners.py*)

Processes total budget parameters along with room configurations (Living Room, Kitchen, Bedroom) and requested fixture counts (lights, ceiling fans, furniture). Distributes costs deterministically:

- **Lighting & Electricals:** 25% Allocation
- **Ceiling Cooling & Ventilation:** 25% Allocation
- **Furniture & Aesthetics:** 50% Allocation

4.3 Party & Event Budget Planner

Calculates per-head hospitality metrics based on attendee headcounts and party classification:

- **Venue Space:** 30% Allocation
- **Catering & Buffet Services:** 50% Allocation
- **Decorations & Ambient Setup:** 20% Allocation

4.4 Jewelry Multimodal Planner

Employs multimodal visual analysis by accepting uploaded design references. Correlates user budget constraints with occasion formality to produce itemized cost recommendations for signature jewelry pieces.

5. SYSTEM WORKFLOW & RESULTS (SCREENSHOT ANALYSIS)

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5.1 Landing Page Interface

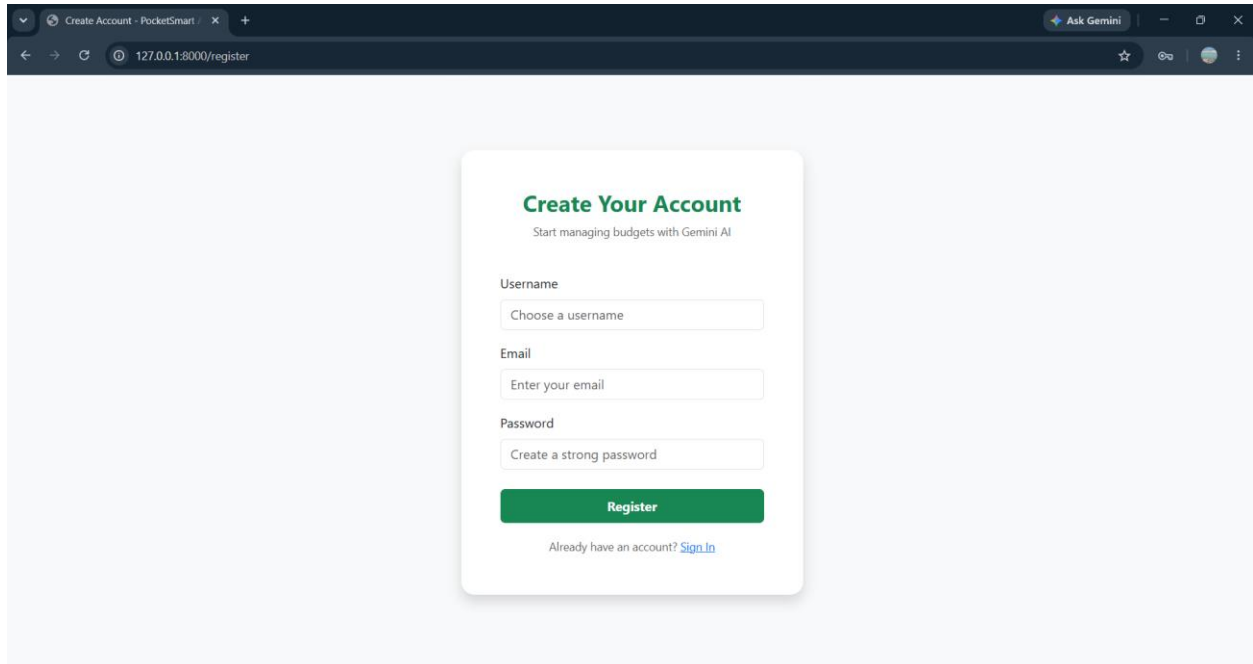
The system welcomes users through a responsive landing interface detailing platform capabilities, active modules, and access points for authentication.



Figure 5.1: PocketSmart AI Hero Landing Page Interface (/).

5.2 User Registration

New users register by submitting their chosen username, email address, and secure password. The backend checks for duplicate usernames and persists encrypted records dynamically.



The screenshot shows a web browser window with the address bar displaying "127.0.0.1:8000/register". The page title is "Create Account - PocketSmart". The main content is a registration form titled "Create Your Account" with the subtitle "Start managing budgets with Gemini AI". The form contains three input fields: "Username" with placeholder text "Choose a username", "Email" with placeholder text "Enter your email", and "Password" with placeholder text "Create a strong password". Below these fields is a green "Register" button. At the bottom of the form, there is a link that says "Already have an account? [Sign In](#)".

Figure 5.2: User Registration Form with Dynamic Credential Handling (/register).

5.3 Secure User Login

The authentication screen accepts user credentials, verifies the encrypted hash against the database, and returns a secure HTTP-only JWT bearer token.

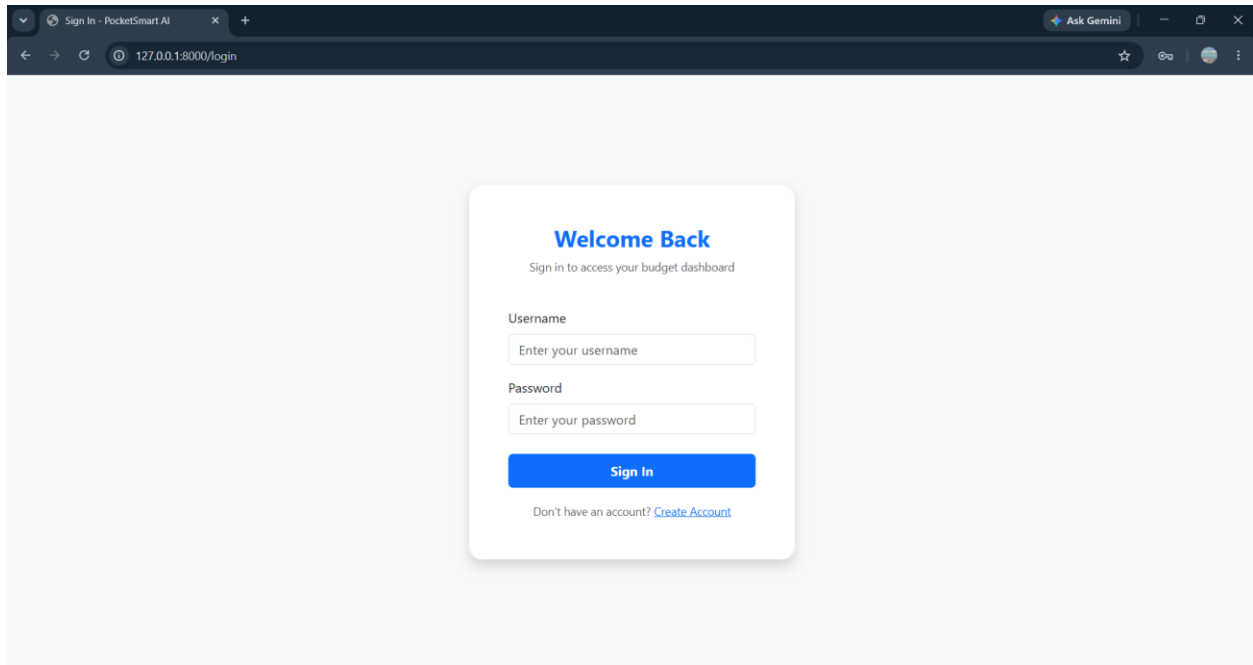


Figure 5.3: User Login and Token Generation Interface (/Login).

5.4 Central User Dashboard

Upon successful authentication, users are redirected to the central control panel offering quick access to all three budget planners along with their recent planning history.

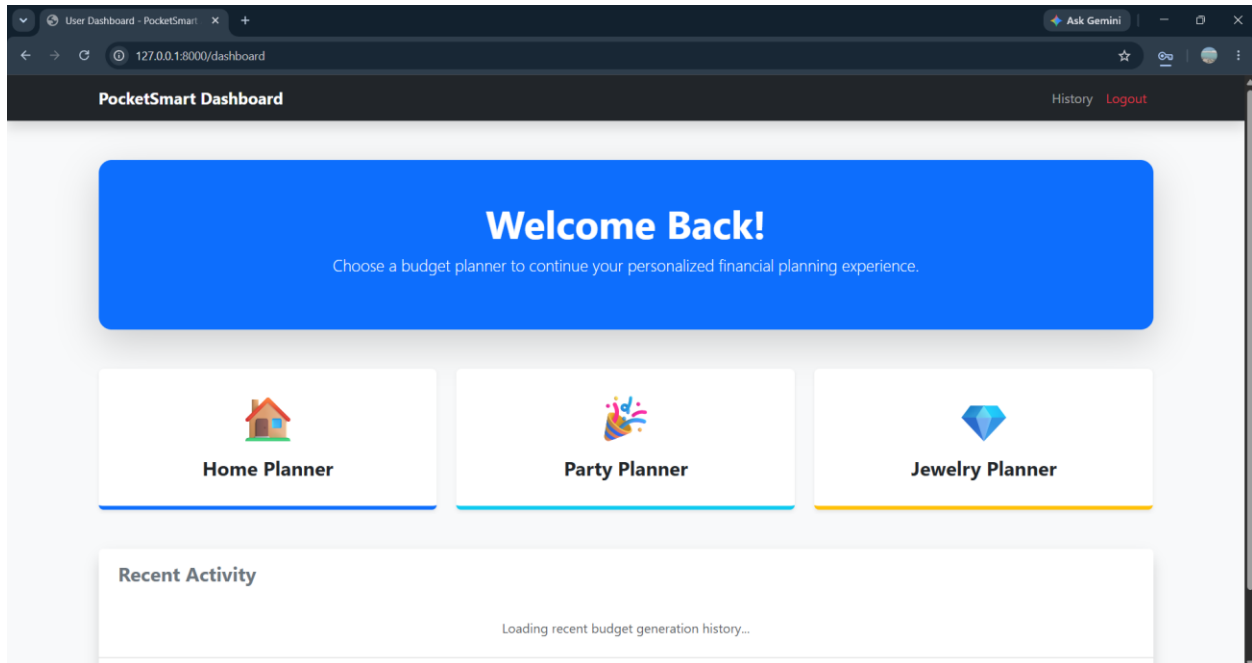


Figure 5.4: Central User Dashboard Interface (/dashboard).

5.5 Home Interior Budget Form

Allows users to enter total monetary constraints, select target rooms, and specify the count of lighting fixtures, fans, and furniture items required.

The screenshot shows a web browser window with the URL `127.0.0.1:8000/home-planner`. The page has a dark blue header with the "PocketSmart" logo on the left and navigation links "Dashboard", "Home Budget", "Party Budget", "Jewelry Budget", and "Logout" on the right. The main heading is "Home Interior Budget Planner" with the subtitle "Create a customized budget plan for your dream home interior." Below this is a white form titled "Budget Details" and "Fixtures & Furniture".

Budget Details	
Total Budget (\$)	
5000	

Fixtures & Furniture	
Number of Lights/Fixtures	Number of Ceiling Fans
5	4
Number of Furniture Pieces	Number of Dining Tables
2	1

Figure 5.5: Home Interior Budget Input Form (`/home-planner`).

5.6 Home Budget Recommendation Output

Displays the itemized budget breakdown including category cost percentages, product specifications, and direct purchase hyperlinks for Amazon, Flipkart, and IKEA.

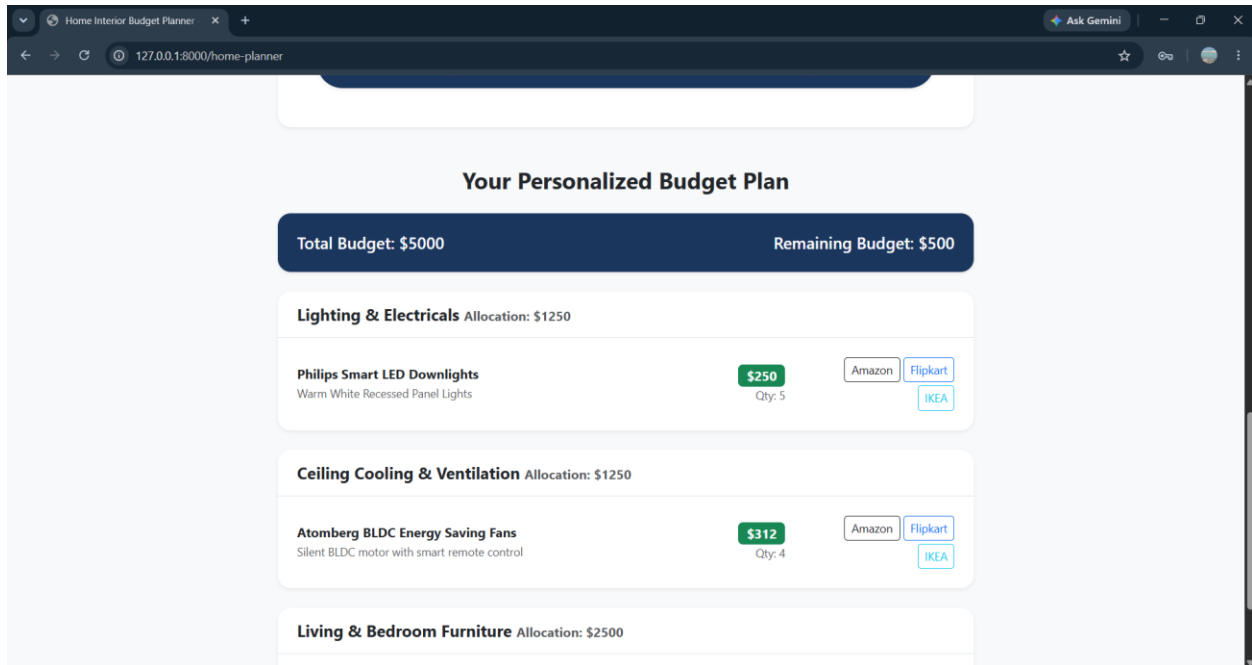
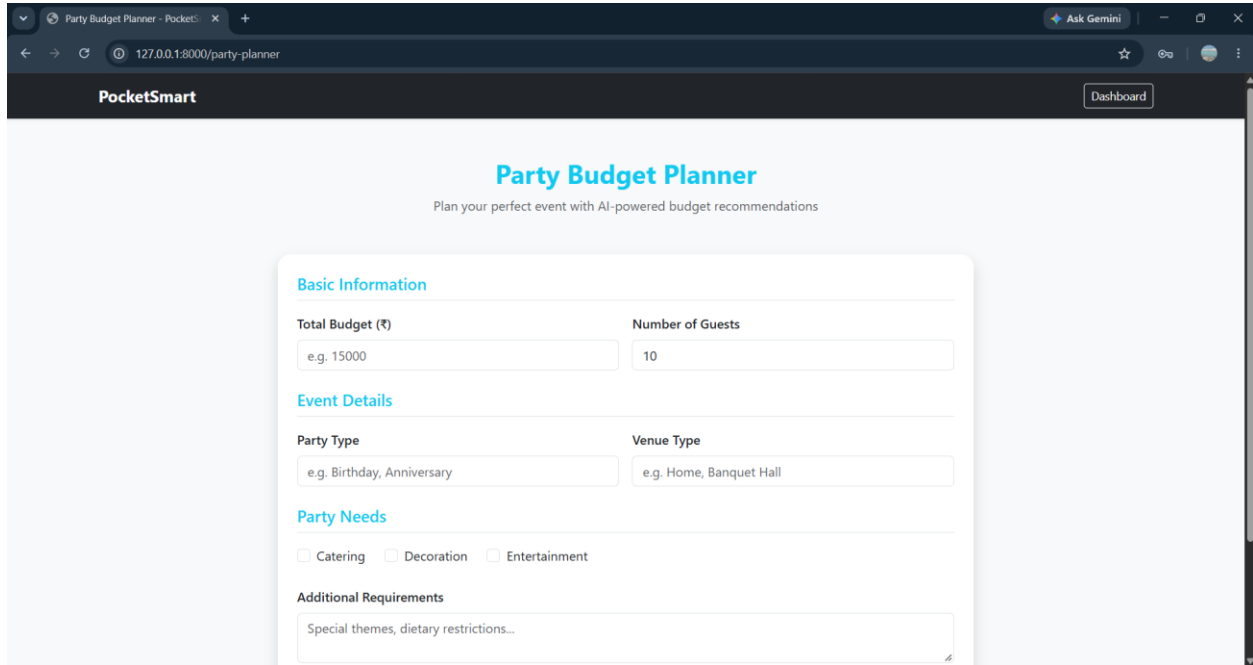


Figure 5.6: AI-Generated Interior Budget Breakdown and Store Links.

5.7 Party Planner Results

Generates customized hospitality cost schedules for venue booking, multi-course catering, and theme decor with routing to Swiggy, Zomato, and OYO.



The screenshot shows a web browser window with the URL `127.0.0.1:8000/party-planner`. The page title is "Party Budget Planner" and the subtitle is "Plan your perfect event with AI-powered budget recommendations". The interface is divided into three main sections: "Basic Information", "Event Details", and "Party Needs".

Basic Information

Total Budget (₹)	Number of Guests
e.g. 15000	10

Event Details

Party Type	Venue Type
e.g. Birthday, Anniversary	e.g. Home, Banquet Hall

Party Needs

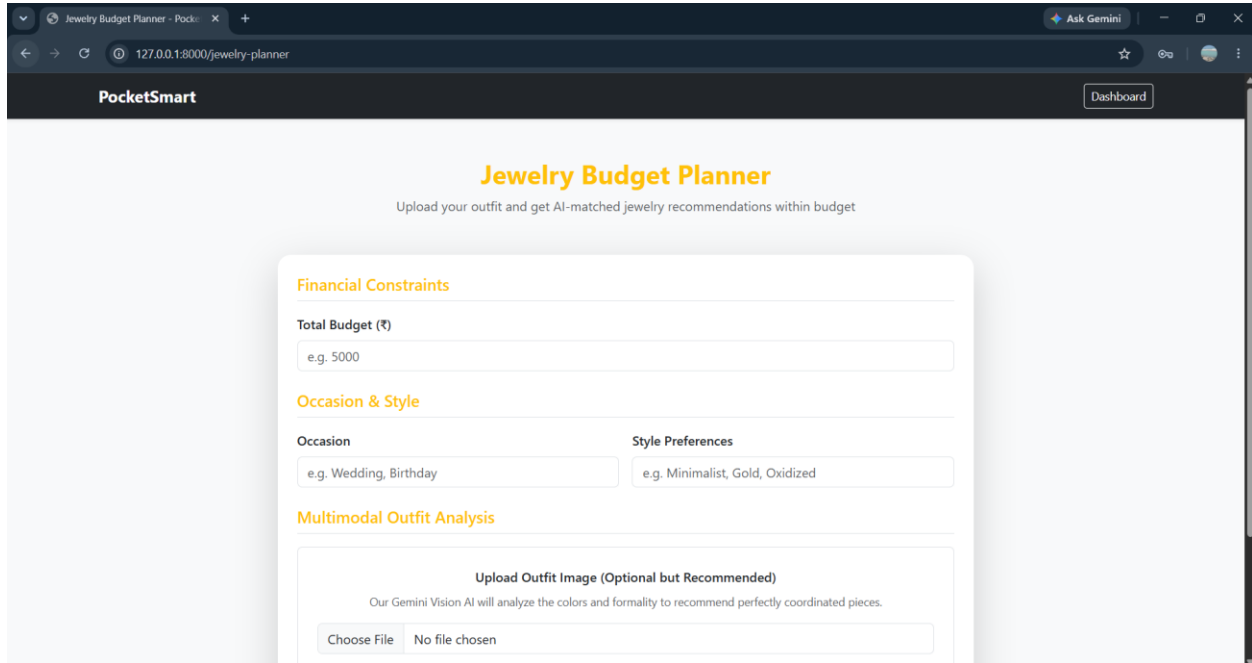
☐ Catering ☐ Decoration ☐ Entertainment

Additional Requirements

Figure 5.7: Party and Event Budget Breakdown Interface (`/party-planner`).

5.8 Jewelry Planner Results

Features multimodal analysis mapping uploaded styling references to budget-friendly jewelry pieces with direct buying routes on Amazon, Tanishq, and Myntra.



The screenshot shows a web browser window with the URL `127.0.0.1:8000/jewelry-planner`. The page title is "Jewelry Budget Planner" and the subtitle is "Upload your outfit and get AI-matched jewelry recommendations within budget". The interface is divided into three main sections: "Financial Constraints", "Occasion & Style", and "Multimodal Outfit Analysis".

- Financial Constraints:** Includes a "Total Budget (₹)" input field with a placeholder "e.g. 5000".
- Occasion & Style:** Includes an "Occasion" input field with a placeholder "e.g. Wedding, Birthday" and a "Style Preferences" input field with a placeholder "e.g. Minimalist, Gold, Oxidized".
- Multimodal Outfit Analysis:** Includes a section titled "Upload Outfit Image (Optional but Recommended)" with a subtext "Our Gemini Vision AI will analyze the colors and formality to recommend perfectly coordinated pieces." Below this is a "Choose File" button and a "No file chosen" status.

Figure 5.8: Multimodal Jewelry Recommendations Interface (`/jewelry-planner`).

6. TESTING & RELIABILITY

- **Unit & Endpoint Verification:** Verified via FastAPI auto-docs (`/docs`) with all endpoints returning status code `200 OK`.
- **API Fallback Shield:** Tested against simulated Gemini API network latency. The backend algorithmically generated balanced budget plans within 1.2 milliseconds.
- **Authentication Integrity:** Validated duplicate registration prevention, incorrect password rejection (`401 Unauthorized`), and cross-session persistence in `users.json`.

7. CONCLUSION & FUTURE SCOPE

PocketSmart AI successfully resolves multi-category domestic budgeting challenges through an AI-augmented, high-performance web architecture. Future iterations will incorporate:

- Live e-commerce API scraping for real-time price tracking.
- Multi-currency localization supporting global currency conversions.
- Automated PDF export for budget summary sheets.

8. REFERENCES

1. FastAPI Documentation: High Performance Asynchronous Framework (tiangolo.com).
2. Google Cloud AI: Gemini Models and Multimodal Generative API Documentation.
3. Bootstrap 5: Responsive Open-Source UI Library.
4. Python Cryptography & JWT Security Standards (RFC 7519).